Regular lambda-definable functions are closed under minimization

The Open Logic Project contributors

openlogic:lambda-calculus/lambda-definability:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author lemma labelled lem:min (971-977); the proof span 980-1008 is the author proof environment that immediately follows it.

Editorial scope. One original result of the Open Logic Project unit lambda-calculus/lambda-definability, counted once: regular lambda-definable functions are closed under minimization. Only the counted lemma and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Lambda definability. A partial function is lambda definable when some lambda term reduces inputs in Church numeral form to the Church numeral of the output whenever the function is defined.. F is the term that lambda defines f, and the constructed H must lambda define g (977-979).
- Fixpoint combinator. A lambda term Y with Y h reducing to a term that self-applies h, giving recursive self-reference in the untyped lambda calculus; the unit uses it as a standard tool.. The search loop is obtained as (Y Search) so Search can call itself with an incremented y (984-986).
- Regular function. A partial function f is regular when for every input the set of y with f at y equal to zero is nonempty.. Regularity guarantees the displayed search reaches the least witness, so the reduction chain ends at the numeral of g's value (991-995).

Lemma 0.1. *If $f(x_1, \ldots, x_k, y)$ is regular and !!lambda definable, then g defined by*

$$g(x_1, \ldots, x_k) = \mu y f(x_1, \ldots, x_k, y) = 0$$

is also !!lambda definable.

Proof. Suppose the lambda term F λ-defines the regular function $f(\vec{x}, y)$. To !!lambda define h we use a search function and a fixpoint combinator:

$$\begin{aligned} &\mathrm{fn}\, Search\, \mathrm{id}\, \lambda[g][\lambda[f\, \vec{x}\, y][\mathrm{fn}\, IsZero(f\, \vec{x}\, y)\, y\, (g\, \vec{x}(\mathrm{fn}\, Succ\, y)]] \\ &\quad H\, \mathrm{id}\, \lambda[\vec{x}][(Y\ \mathrm{fn}\, Search) F\, \vec{x}\, \underline{0}], \end{aligned}$$

where Y is any fixpoint combinator. Informally speaking, $\mathrm{fn}\, Search$ is a self-referencing function: starting with y, test whether $f\,\vec{x}\,y$ is zero: if so, return y, otherwise call itself with $\mathrm{fn}\, Succ\, y$. Thus $(Y\ \mathrm{fn}\, Search)F\underline{n_1}\ldots\underline{n_k}\,\underline{0}$ returns the least m for which $f(n_1,\ldots,n_k,m)=0$.

Specifically, observe that

$$(Y\ \mathrm{fn}\, Search)F\underline{n_1}\ldots\underline{n_k}\,\underline{m}\to\underline{m}$$

if $f(n_1,\ldots,n_k,m)=0$, or

$$\to(Y\ \mathrm{fn}\, Search)F\,\underline{n_1}\ldots\underline{n_k}\,\underline{m+1}$$

otherwise. Since f is regular, $f(n_1,\ldots,n_k,y)=0$ for some y, and so

$$(Y\ \mathrm{fn}\, Search)F\underline{n_1}\ldots\underline{n_k}\,\underline{0}\to\underline{h(n_1,\ldots,n_k)}.$$

□

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